Scientific Management as Work System Redesign (M1.4)

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Scientific Management as Work System Redesign

- Emerged at the intersection of industrial transformation and the Scientific Revolution
- Extended scientific reasoning into organized human work
- Positioned management as a “true science” (Taylor, 1911)
- Core idea: Work itself can be systematically analyzed and redesigned

The Scientific Spirit: From Enlightenment to Industry

- Scientific Revolution established:
  - Systematic observation
  - Precise measurement
  - Quantification of variation
  - Predictive control
- Machinery became scientific; human work remained rule-of-thumb
- Taylor applied scientific method to factory work (Taylor, 1911)

Science?

I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind; it may be the beginning of knowledge, but you have scarecely, in your thoughts, advanced to the stage of Science, whatever the matter may be.

— William Thomson(in 1892, Thomson, then one of Britain's foremost scientists, was raised to the peerage as Lord Kelvin).

The Rise of Measurement and Standardization

Royal Observatory Greenwich
(time standardization)

British Yard / standardized length measures
(measurement standardization)

Frederick W. Taylor: Engineer and System Reformer

Frederick W. Taylor: Engineer and System Reformer

- Began as machinist apprentice → Chief engineer
- Observed:
  - No standardized methods
  - Tacit worker knowledge
  - Informal supervision
  - Output restriction
- Diagnosed inefficiency as systemic disorder
- Treated work as decomposable, measurable, optimizable

Soldiering and System Misalignment

- Natural soldiering: conserving effort
- Systematic soldiering: deliberate output restriction (Taylor, 1911)
- Causes:
  - Worker mistrust
  - Fear of wage cuts
  - Lack of objective standards
- Root problem: flawed work organization

The Four Principles of Scientific Management

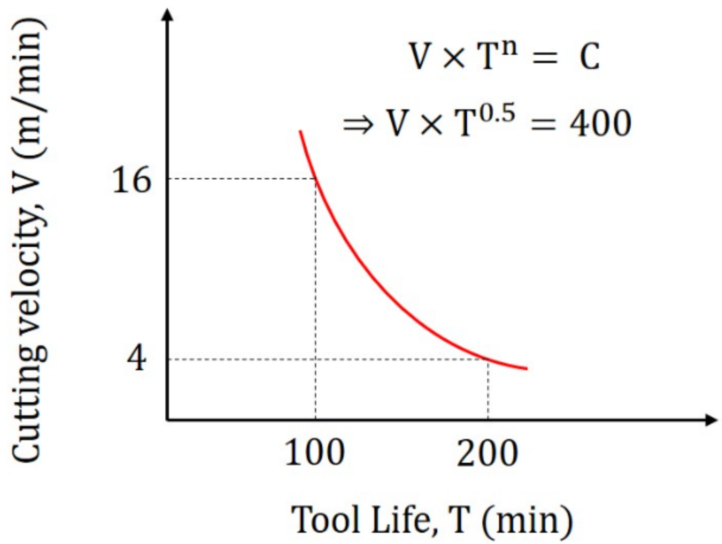
1. **Develop a science for each element of work**
   - **Time study, standardization, tool optimization**
2. **Scientifically select and train workers**
3. **Promote cooperation through aligned incentives**
4. **Divide planning and execution responsibilities**

→ Systematic redesign of organizational structure

Cutting Tool Experiments

- Conducted controlled experiments on:
  - Tool materials
  - Cutting speeds
  - Feed rates
- Derived mathematical machining relationships (Taylor, 1907)
- Demonstrated experimental engineering applied to production

Taylor's tool life equation



Taylor's tool life equation $VT^n = C$, models the relationship between cutting speed (V) and tool life (T) in machining, developed by Frederick Taylor in 1907. It states that tool life decreases as cutting speed increases, where n is the tool life exponent (tool material dependent) and C is a constant based on cutting conditions.

The Pig Iron Case (Bethlehem Steel, 1899)

- Initial condition (Taylor, 1911):
  - Workers manually loaded pig iron (≈92 lbs each)
  - Average output ≈ 12.5 tons per worker per day
  - Paid day wages
  - No standardized method
  - Worker self-pacing
- Taylor’s intervention:
  - Scientifically selected a “first-class worker” (“Schmidt” case)
  - Designed strict work–rest cycles
  - Supervisors instructed exact timing
  - Introduced differential piece-rate system
- Reported result (Taylor, 1911):
  - Output increased to ≈ 47–48 tons per day
  - Wage increased ≈ 60%
  - Fewer workers needed

The Pig Iron Case (Bethlehem Steel, 1899)

- Key System Changes
  - Task pacing externally structured
  - Rest scientifically scheduled
  - Incentives aligned with target output
  - Planning separated from execution
  - Work transformed from self-directed → system-directed

The Shovel Studies

- Problem:
  Different materials (coal, ore, coke) required different loads.
  Workers used identical shovel sizes.
- Taylor's experiment:
  - Determined optimal shovel load ≈ 21 pounds per scoop
  - Designed different shovel sizes for different materials
  - Standardized tools
  - Centralized tool storage and issuance
- Impact:
  - Reduced fatigue
  - Increased output
  - Reduced unnecessary motion
  - Improved control over variability

The Ball Bearing Inspection Study

- Context
  - Bicycle ball bearing factory
  - Inspection of tiny steel balls
  - Task required:
    - Sustained visual concentration
    - High precision
    - Continuous sorting and classification
  - Workforce: ~120 young female inspectors
  - Workday: 10.5 hours (plus Saturday half-day)
  - Paid by day wages

The Ball Bearing Inspection Study

- Taylor concluded:
  - The job required intense attention
  - Fatigue reduced accuracy
  - Workers were present 10.5 hours
  - But effective productive work ≈ **8 hours**
- He recognized:

  Long hours ≠ High productivity

  Cognitive fatigue limits performance
- This is one of the earliest industrial recognitions of cognitive load

The Ball Bearing Inspection Study

- Taylor introduced:
  - Reduced working hours (10.5 → ~8.5 hours)
  - Scheduled rest periods
  - Careful worker selection
  - Increased wages (≈ 80–100% increase)
  - Performance-based incentives
- He removed workers who could not maintain standards.
- System changes:
  - Fatigue management
  - Scientific rest structuring
  - Output measurement
  - Incentive alignment

The Ball Bearing Inspection Study

- According to Taylor:
  - Workforce reduced: ~120 → ~35 inspectors
  - Output increased dramatically
    (Taylor reports increase from approx. 5 million to 17 million inspected units)
  - Accuracy improved (~2/3 improvement in precision)
  - Wages increased substantially
  - Two paid holidays per month introduced
- Taylor emphasized:

  Higher productivity AND higher wages AND shorter hours
- This was presented as a win–win system redesign.

The Ball Bearing Inspection Study

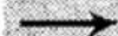
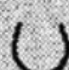
- Historical importance:
  - Unlike the pig iron case, this was not heavy labor.
  - It was:
    - Cognitive work
    - Attention-intensive
    - Precision-based
  - Key contribution:
    - Early recognition of fatigue as a system variable
    - Early integration of:
      - Physiology
      - Psychology
      - Productivity
    - Precursor to modern:
      - Human factors
      - Ergonomics
      - Cognitive workload management

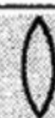
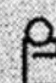
The Gilbreths and Motion Study

- Extended time study into motion study
- Introduced:
  - Micromotion analysis
  - Process charts
  - “Therbligs” (Gilbreth & Gilbreth, 1917)
- Bricklaying experiments dramatically improved efficiency
- Emphasized fatigue reduction and motion economy
- Foundations of modern ergonomics

The Gilbreths and Motion Study

Therbligs

Abbre-viation	Symbol	Name of symbol
Sh		SEARCH
F		FIND
St		SELECT
G		GRASP
TL		TRANSPORT LOADED
P		POSITION
A		ASSEMBLE
U		USE
DA		DISASSEMBLE

Abbre-viation	Symbol	Name of symbol
I		INSPECT
PP		PRE-POSITION
RL		RELEASE LOAD
TE		TRANSPORT EMPTY
R		REST FOR OVERCOMING FATIGUE
UD		UNAVOIDABLE DELAY
AD		AVOIDABLE DELAY
Pn		PLAN
H		HOLD

Bricklaying Revolution (Efficiency via Redesign)

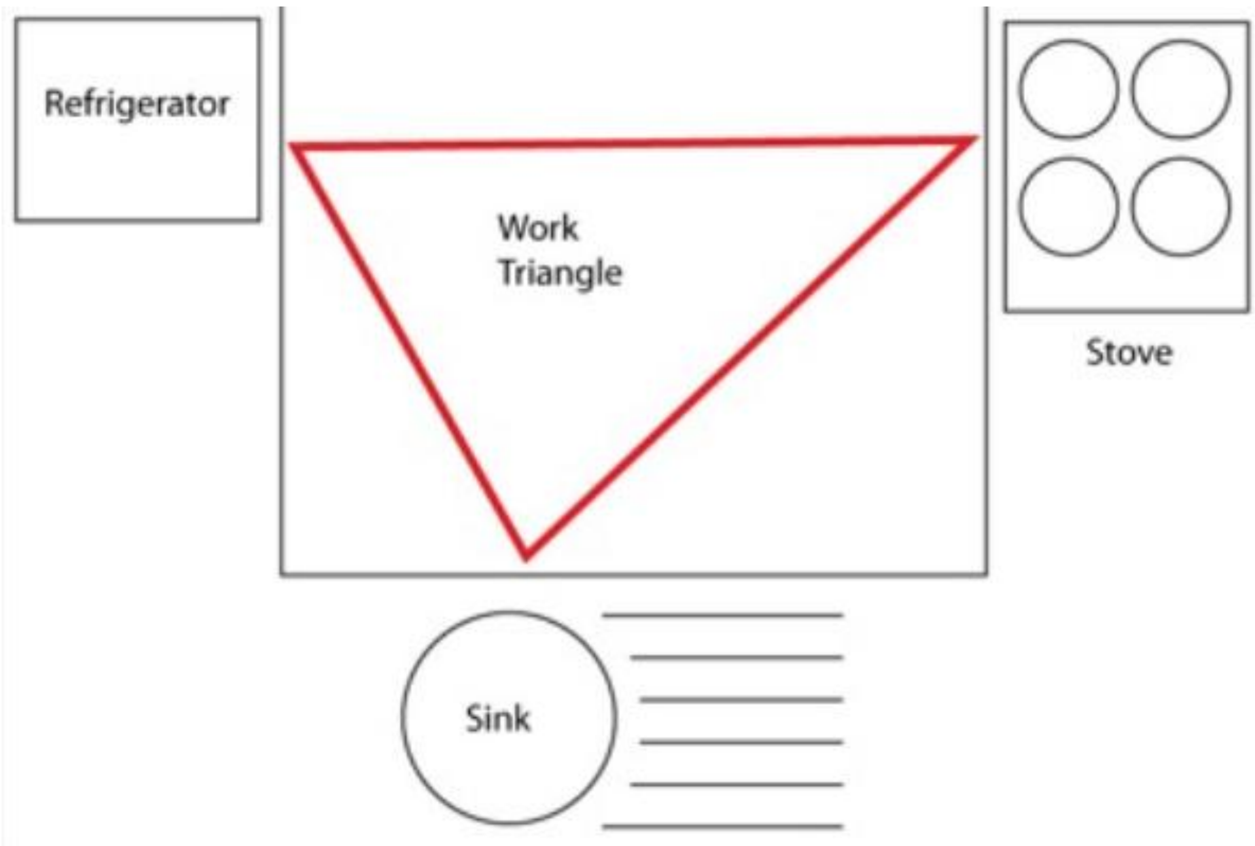
- Gilbreth improved bricklaying by redesigning the work method + workstation
- Key changes:
  - Adjusted scaffold height
  - Positioned bricks at waist level
  - Reduced bending
  - Eliminated unnecessary motions
- Reported productivity gain: ~120 bricks/hour → 300+ bricks/hour
- Teaching point: productivity gains often come from motion elimination, not “working harder”

Household & Kitchen Work as Work Systems

- Lillian Gilbreth applied work design principles to **household work**
- Contributions:
  - Studied kitchen workflows
  - Analyzed reach distances and storage placement
  - Influenced efficiency-focused kitchen layout thinking
- Often linked to ideas later called the **“kitchen work triangle”** (stove–sink–refrigerator)
- Also associated with practical improvements (e.g., can opener design, pedal bins, shelf layout concepts)

Household & Kitchen Work as Work Systems

- The “kitchen work triangle”



Lillian Gilbreth: Major Historical Significance

- Earned a PhD in psychology (Brown University, 1915)
  - Dissertation: *The Psychology of Management*
- Widely recognized as a pioneer in:
  - Industrial psychology
  - Human-centered engineering
  - Early women leaders in engineering/management history
- Often referred to as “The First Lady of Engineering”

Psychology + Engineering Integration

- Taylor emphasized efficiency and structure
- Lillian pushed scientific management toward human-centered concerns:
  - Worker satisfaction
  - Fatigue reduction
  - Motivation
  - Psychological experience of work
- Key message: redesigning work isn’t only speed/throughput—it's also:
  - Information
  - Feedback
  - Responsibility
  - Human capability limits and well-being

Cultural Impact: *Cheaper by the Dozen*

- The Gilbreths had **12 children**
- Family story popularized via *Cheaper by the Dozen* (written by their children)
- Shows motion-study “efficiency thinking” applied to everyday life

Cultural Impact: *Cheaper by the Dozen*

Scientific Management as Work System Engineering

- Redesigned system components:
  - Tasks → standardized sequences
  - Participants → scientific selection
  - Tools → optimized design
  - Information → centralized planning
  - Feedback → measurement systems
  - Incentives → performance-based pay
  - Authority → planning vs. execution

→ Early structured work system engineering

Critiques and Evolution

- Critiques:
  - Deskilling, reduced autonomy (Braverman, 1974)
- Sociotechnical theory:
  - Joint optimization of social and technical systems
  - Emphasis on autonomy and adaptability
- Modern design integrates:
  - Measurement
  - Ergonomics
  - Sociotechnical balance

Mini-Summary

- Scientific management:
  - Applied scientific method to work
  - Formalized time study and standardization
  - Introduced planning structures and feedback control
  - Extended by Gilbreths’ motion study
  - Initiated systematic work system engineering
  - Powerful yet incomplete → evolved into sociotechnical and ergonomic design